Pressione di radiazione

2/7/21 11:30 AM

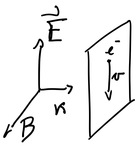
Pressione di radiazione: forza per unità di superficie esercitata da un'onda sulla superficie di un mezzo ortogonale alla dire prop

Per 1 carica e Σ

$$\vec{F}_q = q\left(\vec{E} + \vec{v}\times\vec{B}\right)$$

$$\frac{d\vec{F}}{dS} = \sigma\left(\underset{①}{\vec{E}} + \underset{②}{v\times\vec{B}}\right)$$

① $\frac{dF_e}{dS} = \sigma\vec{E}$ mette in moto gli e^- con $v \parallel \Sigma \parallel \vec{E}$



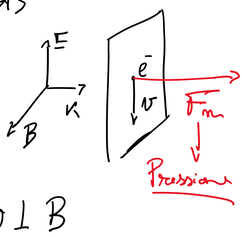
$\Downarrow$

assorbimento di energia

$$\frac{P_{ass}}{dS} = \frac{d\vec{F}}{dS}\cdot\vec{v} = \sigma E v$$

$\bar{I} = \sigma\bar{E}\bar{v}$ intensità media ass

② $\frac{d\vec{F}_m}{dS} = \sigma\vec{v}\times\vec{B}$



$v \perp B$

$$\frac{dF_m}{dS} = \sigma v B$$

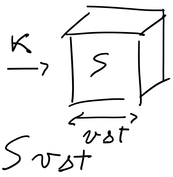
$$\frac{d\bar{F}_m}{dS} = \sigma\bar{v}\bar{B} = \sigma\left(\frac{E}{v}\bar{v}\right) = \sigma\frac{\bar{I}}{v}$$

$$p_{em} := \frac{d\bar{F}_m}{dS} = \frac{\bar{I}}{v}$$

$\exists\ \vec{g}$: quantità di moto / m^3

$$\vec{g} = \frac{\vec{P}}{v^2} = \varepsilon\vec{E}\times\vec{B}$$ se piana $\bar{g} = \frac{\bar{I}}{v^2} = \frac{\bar{u}}{v}$

MEZZO TOTALMENTE ASS.



$$G = \bar{g}\,S\,v\,\Delta t$$

$$I = \int_0^{\Delta t} F\,dt = \Delta G$$

$$F\Delta t = \Delta G = g\,S\,v\,\Delta t$$

$$p_m = \bar{g}v = \bar{u} = \frac{\bar{I}}{v}$$

Se TOTALMENTE RIFL

$$F\Delta t = \Delta G = 2\bar{g}\,S\,v\,\Delta t$$

$\downarrow$ $\times 2$

torna indietro l'onda

$$p_{em} = 2\bar{g}v = 2\bar{u} = \frac{2\bar{I}}{v}$$